Signal Conditioning

By

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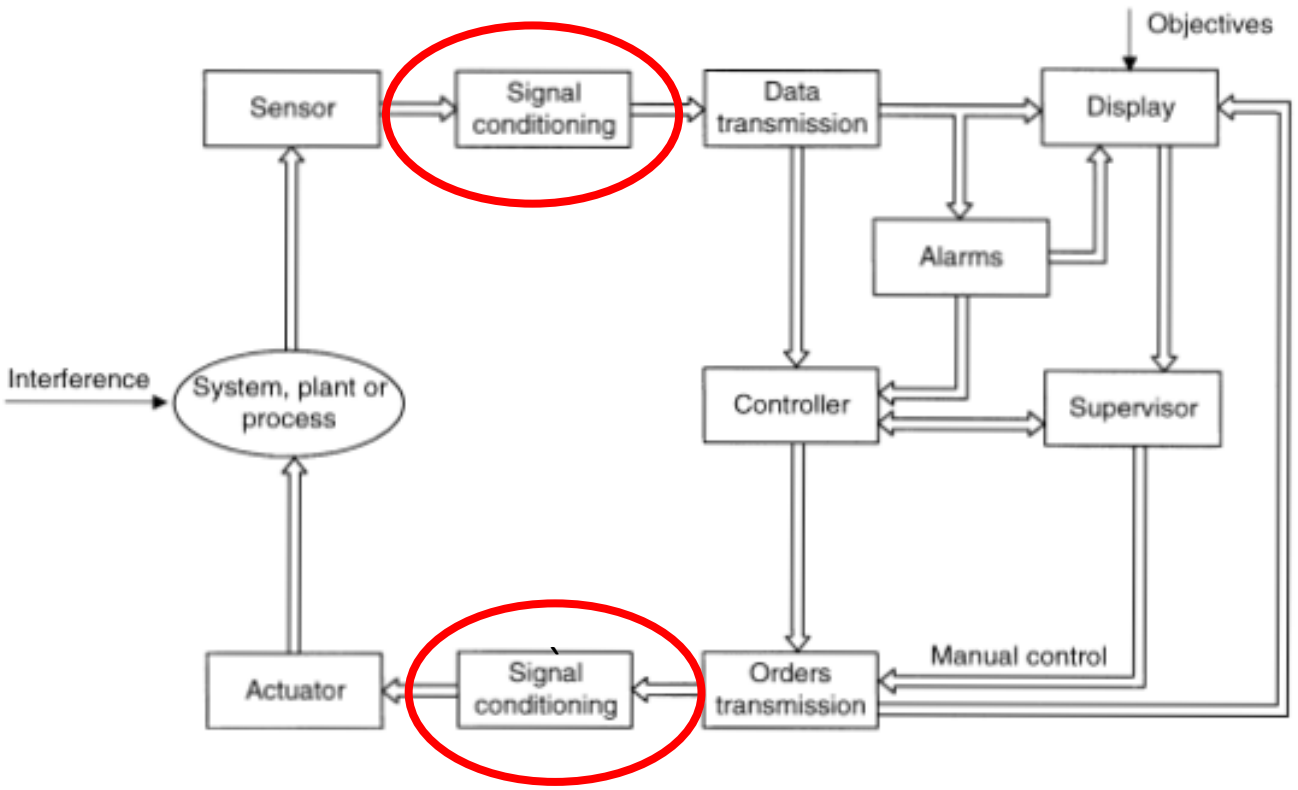
Course Objectives

- Get basic theoretical and practical skills for design of the interface circuits for processing signals from/to sensors/actuators.
- The treatment of the data from the digital and smart sensors.
- Get practical experience with software for theoretical design and simulation of these circuits, and hardware and software resources for processing and digitization of analog signals from sensors.



Course Syllabus

Lecture #	Topics
1	Introduction • Signal Conditioning [What – Why – How]. • Overview of Basic Electrical Components and Signals
2	Overview of Proteus (Electronic Design Automation Software) Overview of Operational Amplifier Basics #1 • Amplifier - Comparator - Follower
3	Overview of Operational Amplifier Basics #2 • Summing - Integration - Differentiation - Voltage/Current conversion - Instrumentation Amplifier
4	Tasks to be done by Signal Conditioning Circuits #1 • Signal Amplification [Level Shifter – Buffer] • Filtering [Analog]
5	Tasks to be done by Signal Conditioning Circuits #2 • Filtering [Digital]. • Isolation - Protection - Linearization – Excitation - Impedance Changing
6	Tasks to be done by Signal Conditioning Circuits #3 • Clipping – Clamping – Multiplexing • Interfacing with µP (ADC)

Course Syllabus

Lecture #	Topics
7	Applications for Sensors and Actuators Interfacing #1 Signal Conditioning for: • Resistive Sensors - Reactance Variation Sensors
8	Applications for Sensors and Actuators Interfacing #2 Signal Conditioning for: • Self-Generating Sensors - Digital Sensors
9	Applications for Sensors and Actuators Interfacing #3 Signal Conditioning for: • Intelligent (Smart) Sensors - Loads Requiring AC Switching or Large Current Flow
10	Experimental Issues • Error compensation - Grounding in mixed-signal systems • Preventing RFI rectification – Shielding – Electrostatic Discharge
11	Data Acquisition • Signal Sources and Measurement Systems - Analog input (A/D) boards • Analog output (D/A) boards - Digital I/O boards - Counter/timer I/O boards
12	Project

Learning Activities and Teaching Methods

- Lectures
- Student Presentations
- Assignments
- Software laboratories
- Project

Assessment methods

- Up to 40 % for the student activities:
  - Assignments ➔ Ensure your understanding of lecture content
  - Presentations ➔ Enhance your research and presentation skills
  - Project ➔ Achieve course aims (Enlighten your Design, Analysis, Simulation, Implementation Skills)
- Up to 60 % for the written exam.

Recommended reading [References]

1. SHEINGOLD, D. H. Transducer Interfacing Handbook - A Guide to Analog Signal Conditioning. Analog Device, Inc., USA, First edition, March 1980. 231 p. ISBN 0 916550-05-2. (EN)
2. KITCHON CH., COUNTS, L. A Designer's Guide to Instrumentation Amplifiers. Analog Devices, Inc., USA, Second Edition, 2004. 108 p.(EN)
3. PALLÁS-ARENY, R., WEBSTER, J. G. Sensors and Signal Conditioning. John Wiley & Sons, Inc., USA, Second edition, 2000. 587 p. ISBN 0 471-33232-1. (EN)
4. HOROWITZ, P., HILL, W. The art of electronics. Cambridge University Press, UK, Second Edition, 1989 (2001). 1125 p. ISBN 0 521-37095-7. (EN)
5. Data Acquisition and Signal Conditioning Course Manual From NI
6. Gómes, C. "An experimental apparatus for signal conditioning circuits development and testing: a teaching experience." (1970).
7. Kester, Walt, ed. *Practical design techniques for sensor signal conditioning*. Analog devices, 1999.
8. Jackson, Leland B. *Digital Filters and Signal Processing: With MATLAB® Exercises*. Springer Science & Business Media, 2013.
9. Fernýndez, Juana, et al. "Digital filter design with Arduino DUE and Matlab." *Information Processing and Control (RPIC), 2015 XVI Workshop on*. IEEE, 2015.

Lecture 1

Introduction

- Signal Conditioning [What – Why – How]
- Overview of Basic Electrical Components and Signals

Lecture 1

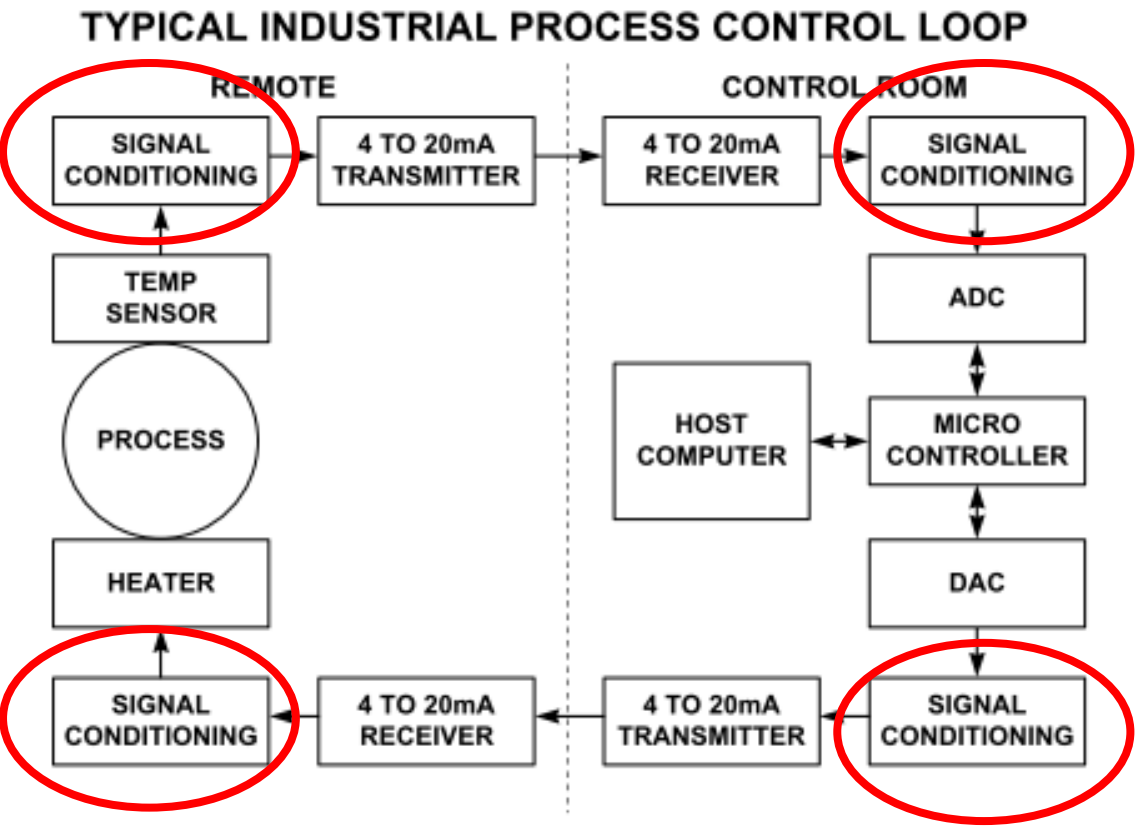
Introduction

- Signal Conditioning [What – Why – How]
- Overview of Basic Electrical Components and Signals

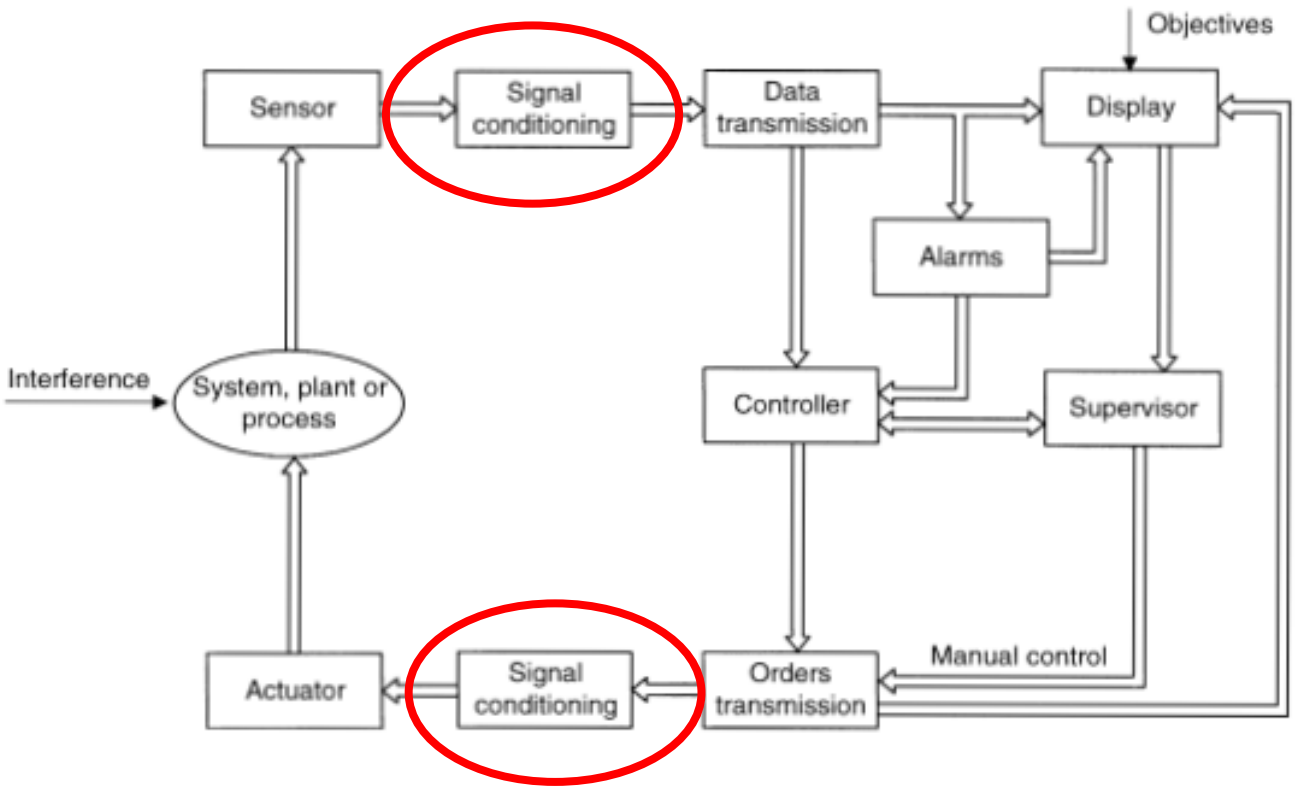
What is “Signal Conditioning”?

- There are many factors which may prevent a signal produced by one device or circuit from being usable by another device or circuit, requiring some intermediate circuitry to bridge the gap.
- This kind of “bridging” function is doing what we call “signal conditioning”.
- Often, the signals outputted by the sensors are not adequate for data acquisition.
- In addition, some sensors required an external excitation or bridge completation to operate.
- The function of the signal conditioning circuits include the following items:
  - Signal amplification
  - Filtering
  - Protection – Linearization
  - Current – voltage change circuits
  - Resistance change circuits
  - Error compensation
  - Interfacing with µP (ADC)

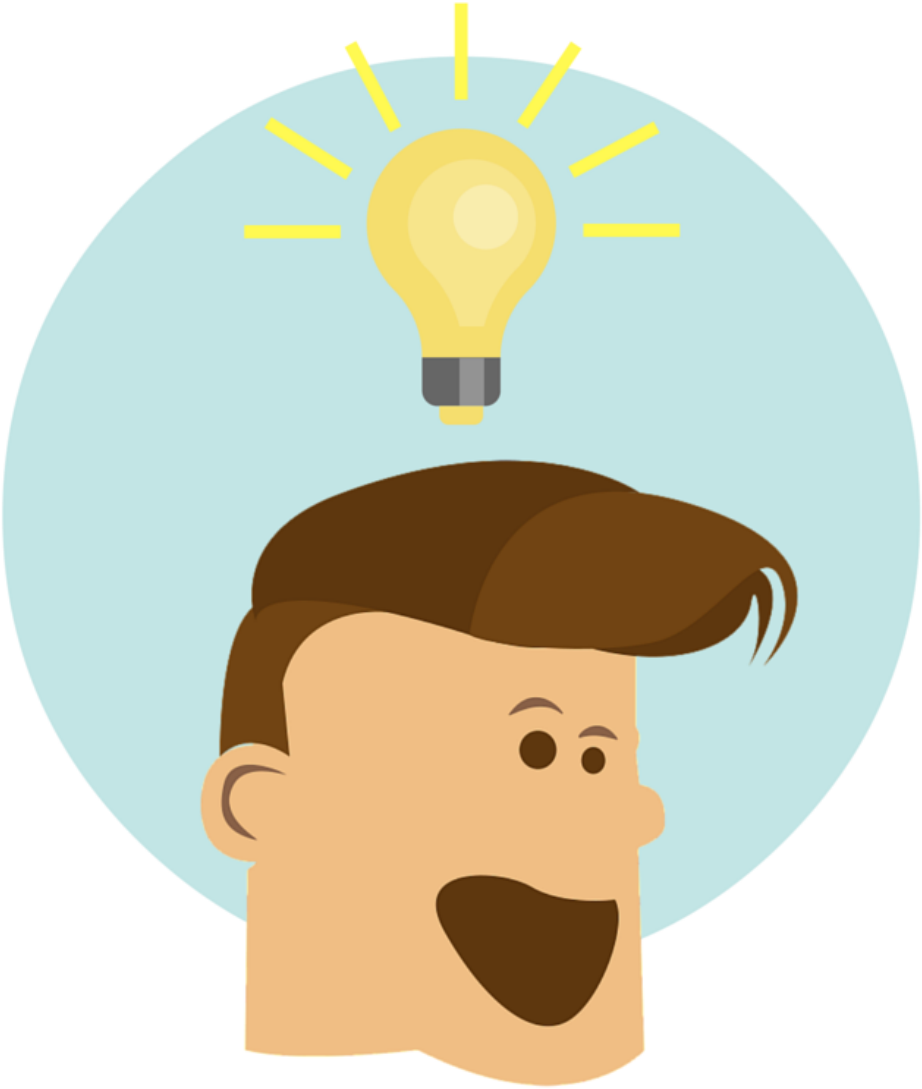
Examples of Real Applications Of SC



Example #1



Example #2



Learning Moments